

Wave guides as an analogous system of 1D optical lattice for anyons

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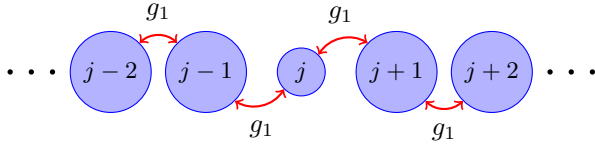
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Keywords: anyons, quantum matter, ultracold atoms

I. INTRODUCTION

Los anyones son ...

II. MAIN IDEA



Para las guías de onda:

- Una guía de onda pose un índice de refracción en el otro núcleo, rodeado por otro de menor índice $n_1 < n_2$
- Para una única guía de onda, los modos admitidos están dados por sus características físicas.
- Un arreglo de guías de onda, el acoplamiento permitirá la transferencia entre los elementos del arreglo.

From an infinite waveguides array with homogeneous optical single mode waveguides and equally spaced along z-axis. This is described by the following differential equation [1]

$$i \frac{dE_j(z)}{dz} = g_1(E_{j-1} + E_{j+1}), \quad j = -\infty, \dots, \infty, \quad (1)$$

From this

$$i \frac{d|\psi(z)\rangle}{dz} = \hat{H}|\psi(z)\rangle, \quad \hat{H} = g_1 (\hat{V}_\infty + \hat{V}_\infty^\dagger), \quad (2)$$

where the operators $\hat{V}_\infty$ and $\hat{V}_\infty^\dagger$ are the infinite counterparts of the London-Susskind-Glogower operators [2, 3], defined as

$$\hat{V}_\infty = \sum_{n=-\infty}^{\infty} |n\rangle\langle n+1|, \quad \hat{V}_\infty^\dagger = \sum_{n=-\infty}^{\infty} |n+1\rangle\langle n|, \quad (3)$$

For an analytical solution we need to satisfy in our Hilbert space the following conditions for the LSG operator

$$[\hat{V}_\infty, \hat{V}_\infty^\dagger] = 0, \quad \hat{V}_\infty \hat{V}_\infty^\dagger = \hat{I},$$

In this framework the initial state is a photon in the n site of an infinite array of waveguides $|\psi(0)\rangle = |n_0\rangle$;

$$|\psi_{n_0}(z)\rangle = \exp \left[-g_1 z \left(i\hat{V}_\infty^\dagger - \frac{1}{i\hat{V}_\infty} \right) \right] |n_0\rangle.$$

Donde podemos identificar la función generadora de las funciones Bessel, para obtener

$$|\psi_{n_0}(z)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2g_1 z) i^n \hat{V}_\infty^{\dagger n} |n_0\rangle.$$

para este modelo podemos hacer una analogía con un modelo de bosones en una red óptica partiendo del operador $\hat{V}_\infty$ y la evolución será escrita en función del tiempo:

$$\hat{V}_\infty = \sum_{n=-\infty}^{\infty} \hat{b}_n^\dagger \hat{b}_{n+1}, \quad \hat{V}_\infty^\dagger = \sum_{n=-\infty}^{\infty} \hat{b}_{n+1}^\dagger \hat{b}_n$$

The operator $\hat{V}_\infty^{\dagger n}$ is equivalent of create a particle in n site in the framework of optical lattices for a subspace of one particle per site:

$$\hat{V}_\infty^{\dagger n} |1_0\rangle = \hat{b}_n^\dagger |0\rangle, \quad (4)$$

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Obtaining the dynamics for a particle in 0-site

$$|\psi_0(t)\rangle = \sum_{n=-\infty}^{\infty} J_n(-2Jt) i^n \hat{b}_n^\dagger |0\rangle. \quad (5)$$

We could obtain a rich dynamics for n-particles system working with a Heisenberg representation for one body operators $\hat{b}^\dagger$ applied in the sites, obtaining in

general the expression:

$$\langle \hat{n}_j \rangle = \sum_{m=-\infty}^{\infty} |J_{m-j}(-2Jt)|^2 \langle \hat{n}_m(0) \rangle \quad (6)$$

$$S(q) = \sum_{l=0}^M |J_l(-2Jt)|^2 - \sum_{l,m=0}^M e^{iq(l-m)} |J_l(-2Jt)|^2 |J_m(-2Jt)|^2$$

III. CONCLUSION

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Appendix: Appendix

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